

U.S. logistics costs and the weight of inventory (2000–2014)

Part 1 · Why is it important? — solution report with data, method, results and charts

Contents

1. Part 1 · Why is it important? — U.S. logistics costs and the weight of inventory (2000–2014)

1. U.S. logistics costs and the weight of inventory (2000–2014)

The U.S. economy held between **\$1.4 and \$2.5 trillion** of business inventory each year in 2000–2014. Holding it cost **19–25 % of its value per year** (the *inventory carrying rate*), making inventory carrying cost roughly one-third of all U.S. logistics spending.

$c = a \times b$	$f = c + d + e$	share of GDP = f / GDP
\$2,496 bn Inventory held in 2014	\$477 bn Carrying cost in 2014	34.0 % Inventory share of logistics cost (avg)
\$1,019 → \$1,433 bn Total logistics cost 2000 → 2014	7.8 % (2009) Lowest logistics share of GDP	3.58 % / yr Inventory value CAGR 2000–2014
0.49 bn (rounding) Max check error $c = a \times b$	0.00 bn Max check error $f = c + d + e$	

U.S. logistics cost composition, 2000–2014

Transportation is the largest component; inventory carrying is a steady ~1/3 of the total

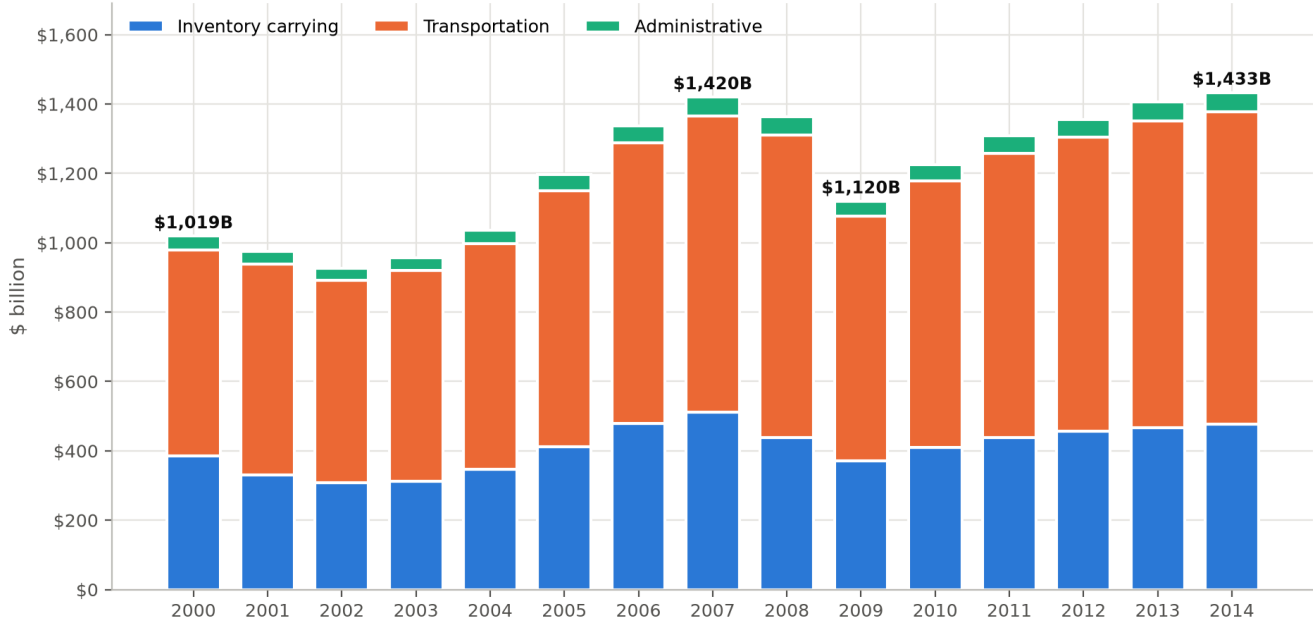


Figure 1.1 — Composition of U.S. logistics costs (\$ billion). Source: lecture slide 5.

Inventory indicators move with the business cycle

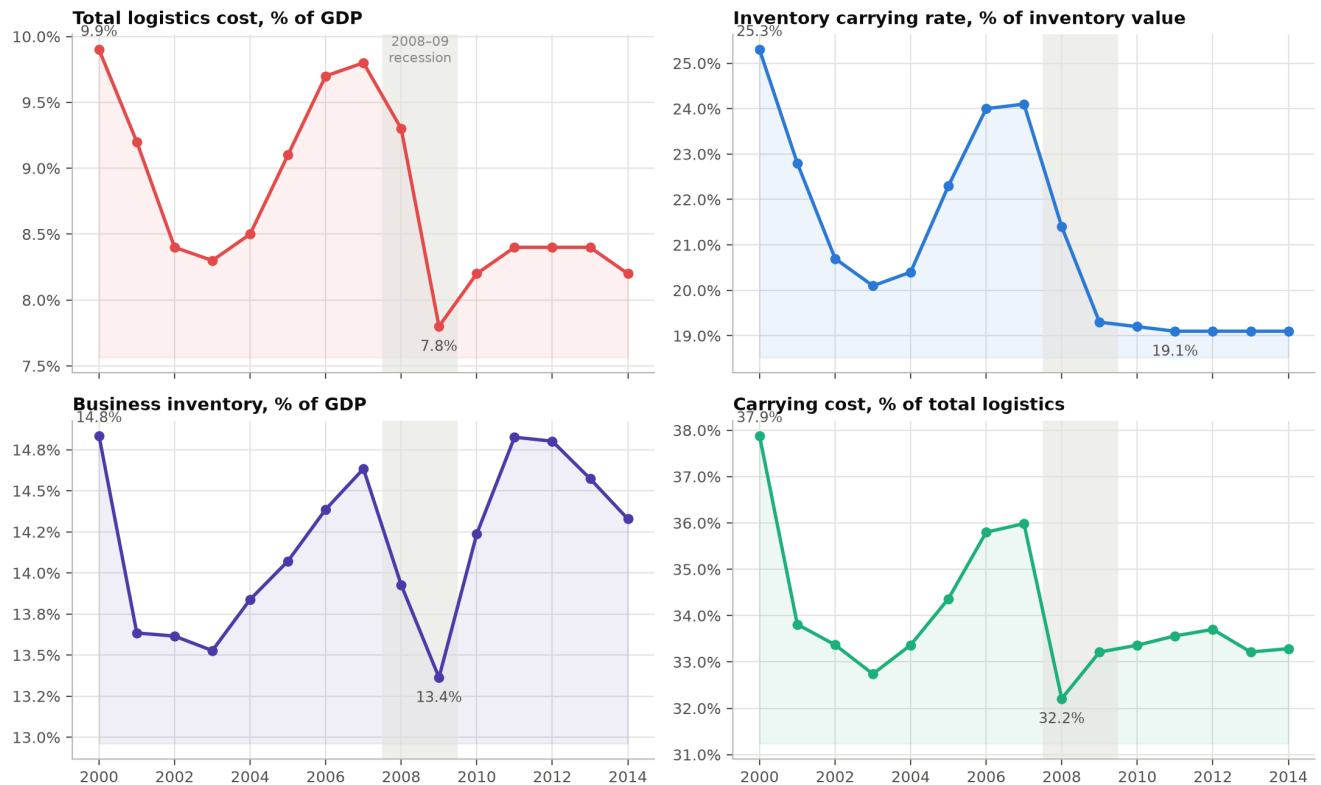


Figure 1.2 — Key ratios derived from the lecture data. The 2008–09 recession (shaded) cut both inventory and logistics intensity.

Growth Indexed to 2000

Inventory value grew ~64 % while its carrying cost grew only ~24 %: cheaper capital lowered the carrying rate

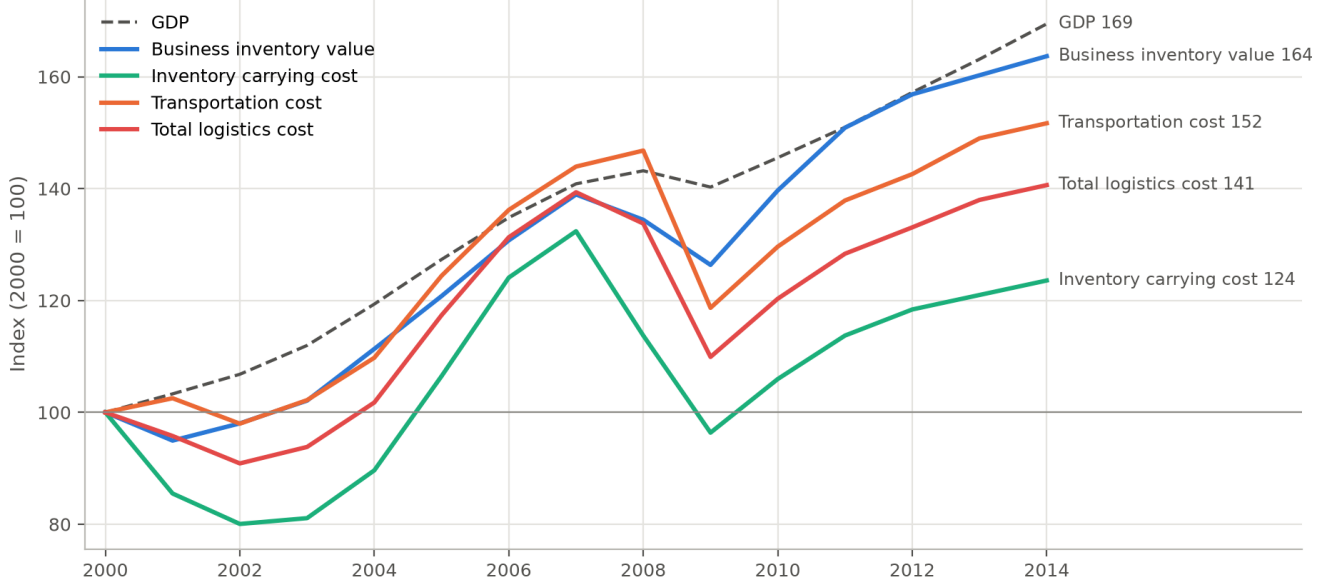


Figure 1.3 — All series indexed to 2000 = 100 so that different magnitudes share one axis.

Lecture data with verification columns ($\text{check_carrying_bn} = a \times b$) and derived shares.

year	gdp bn	business inventory bn	inventory carrying rate pct	inventory carrying cost bn	check carrying bn	transportation cost bn	administrative cost bn	total logistics cost bn	logistics pct gdp	inv share of logistics pct
2000	10,280	1,525	25.3	386	385.8	594	39	1,019	9.9	37.9
2001	10,620	1,448	22.8	330	330.1	609	37	976	9.2	33.8
2002	10,980	1,495	20.7	309	309.5	582	35	926	8.4	33.4
2003	11,510	1,557	20.1	313	313.0	607	36	956	8.3	32.7
2004	12,270	1,698	20.4	346	346.4	652	39	1,037	8.5	33.4
2005	13,090	1,842	22.3	411	410.8	739	46	1,196	9.1	34.4
2006	13,860	1,994	24.0	479	478.6	809	50	1,338	9.7	35.8
2007	14,480	2,119	24.1	511	510.7	855	54	1,420	9.8	36.0
2008	14,720	2,050	21.4	439	438.7	872	52	1,363	9.3	32.2
2009	14,420	1,927	19.3	372	371.9	705	43	1,120	7.8	33.2
2010	14,960	2,130	19.2	409	409.0	770	47	1,226	8.2	33.4
2011	15,520	2,301	19.1	439	439.5	819	50	1,308	8.4	33.6
2012	16,160	2,392	19.1	457	456.9	847	52	1,356	8.4	33.7
2013	16,770	2,444	19.1	467	466.8	885	54	1,406	8.4	33.2
2014	17,420	2,496	19.1	477	476.7	901	55	1,433	8.2	33.3

Takeaway. Inventory is not just a warehouse issue — at a national scale it represents hundreds of billions of dollars of cost each year. Even a one-point reduction in the carrying rate on \$2.5 trillion of inventory saves about **\$25 billion per year**.